ENGR 111
Introduction to Engineering
Class #6

Keys to success in engineering study

Becoming an engineering student

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

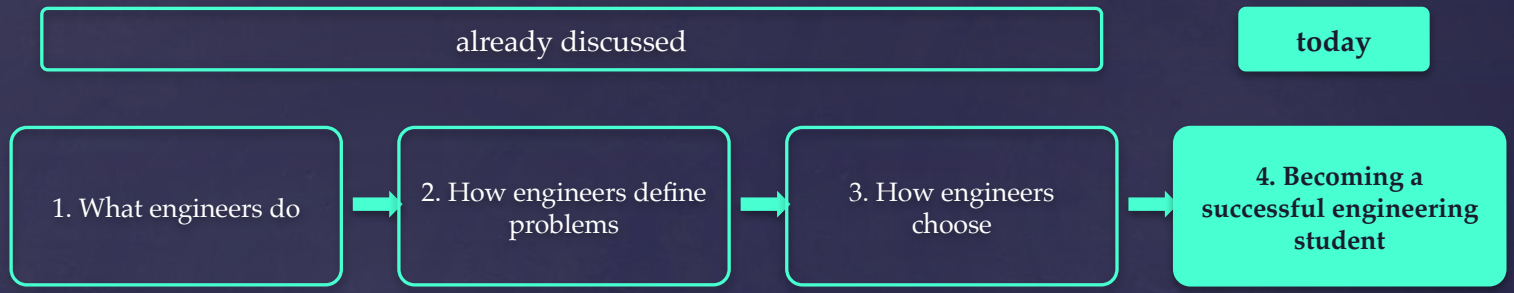
My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

Becoming an engineering student

The point: get into the frame of mind and habits the work requires.



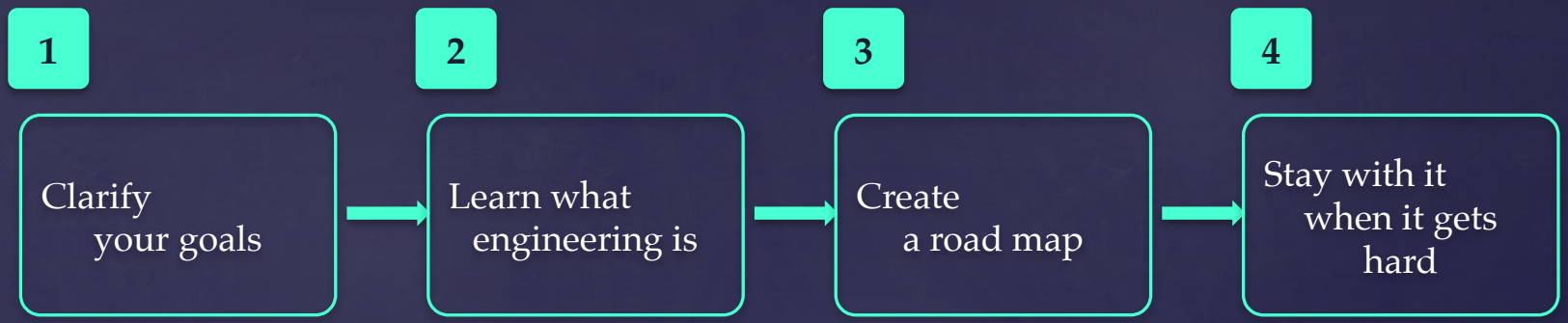
Today. Engineering education is demanding. How do you keep doing hard technical work every week, even when life is crowded?

Later. Learning science, transfer pathways, MECOP, internships, resumes, and career choices.

Tools for today. Effort, approach, attitude, the 60-Hour Rule, and moving the ball forward.

The textbook gives a road map

Four practical strategies you can use to strengthen your commitment to success in engineering study



Clarify the goal. Know why the degree matters to you and what it would change.

Learn the field. The more you understand engineering, the easier it is to want the work.

Map the route. Lay out the courses and steps one term at a time.

Keep going. Expect adversity; pause, reassess, and begin again.

Treat study as a design problem

Start with what you know now, then observe and adjust as the system changes.

Design language	Student version
Problem	How do I succeed in engineering study?
Known parameters	Current classes, math level, work, commute, family.
Constraints	Only so many usable hours and so much attention.
Controls	Course load, schedule, study location, help, habits.
Feedback	Observe, reassess, adjust, and begin again.

The first version will be imperfect. That is normal engineering.
Observe what happens, change the parameters you can change, and "just begin again".

Start where you are

Preparation means the starting conditions: math background, study habits, time, support, health.

Starting behind

Some students start behind and still finish.

They change the system around the work.

The starting point is real. It is not the whole story.

Starting strong

Some students arrive strong and still fail.

They assume college will work like high school.

Overconfidence hides the need to change.

If things get off track: pause, assess the facts, and ”just begin again".

Effort, approach, attitude

Effort. Putting in time and mental energy.

Approach. How you use that time: spacing work out, preparing early, studying with people, asking questions, checking your method.

Attitude. How you respond when the work is difficult, slow, or not going well.

Classify these (do they fall under "effort", "approach" or "attitude"

1. Read the required reading before class
2. Study with people
3. Ask a question
4. Just begin again
5. Put the phone away
6. Think for yourself instead of first resorting to AI
7. Respect your classmates
8. Exercise, eat right, and get sleep
9. Be curious
10. Making studying a true priority

The categories may overlap.

Move the ball forward (down the field)

Make a little progress on something every day.

Small counts. Read a little. Start one problem. Sketch one version. Send one email.

Start early. Avoids panic. Figure out what you need or don't understand early before it becomes a problem.

Keep the content warm. Content or a class you touch often is easier to re-enter than one you ignored for a week.

Missing it should feel wrong. Like the gym: the habit is working when skipping starts to feel unusual.

Small progress compounds. Achieve big outcomes with small, consistent, compounding steps.





The 60-Hour Rule

The theory: you have about sixty work-hours in a week. Everything draws from that budget.

Academic work counts. Class time plus study time. A 3-credit course usually means about 9 hours each week.

Paid work counts. A job is not outside the budget. It spends the same hours.

Commuting counts. The textbook includes it directly.

Other responsibilities count too. Child care, elder care, and major family obligations use real work-hours.

Use it to choose a load

ENGR-111 is 3 credits, so budget roughly 9 hours per week for this course.

The book's estimate. Each credit is about 1 hour in class plus 2 hours outside class.

A 12-credit term. About 36 academic hours before paid work, commuting, or care responsibilities.

A 16-credit term. About 48 academic hours before anything else.

Paid work + commute + care	Course load the book suggests
about 10 hrs/wk	full load
about 20 hrs/wk	12 credits
about 40 hrs/wk	8 credits

Change the system

A few controls you can actually move.

Show up. Do not miss class casually. Missing once creates debt you still have to pay.

Sit in the front. You are using your precious time to show up anyways, might as well sit in the front and learn what you can.

Start early. Early work gives you time to get stuck, ask, recover, and revise.

Get involved on purpose. Classmates, clubs, project teams, internships, MECOP. Details come later, but involvement is part of the route.

Revise the plan. If the current setup only works during a perfect week, it does not work.

Put down the phone. Avoid distractions/multitasking when in class or working. This is maybe the most difficult one.

What are some "front-row" behaviors and other useful tactics for students in a Zoom based online class?